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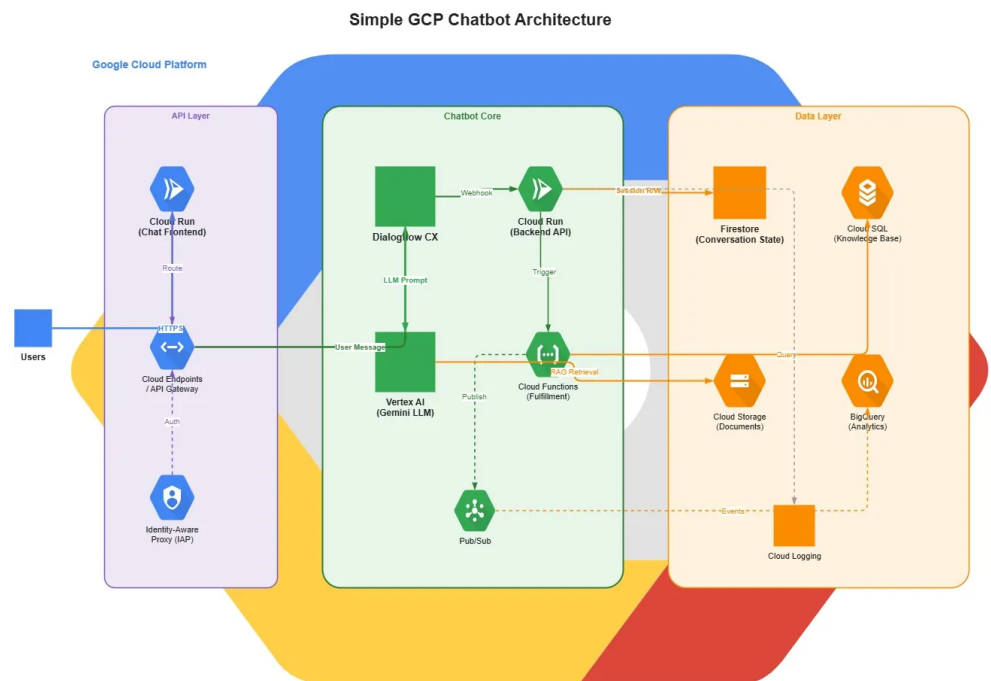
Configuring the Draw.io MCP Server in VS Code for AI-Generated Diagrams



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AI generated architecture within 30 seconds

AI assistants like GitHub Copilot, Claude, and Cline are

becoming powerful development partners. One of the most interesting capabilities they now offer is **AI-generated architecture diagrams directly inside your editor.**

Using the **Draw.io MCP (Model Context Protocol) server**, you can instruct an AI assistant to create flowcharts, architecture diagrams, and system designs automatically.

This article walks through how to configure the **Draw.io MCP server in VS Code**, how to troubleshoot common issues, and how to generate diagrams using AI prompts.

Why Use Draw.io MCP?

Traditionally, creating architecture diagrams meant manually dragging shapes in tools like draw.io or Lucidchart.

With MCP integration:

AI generates diagrams from natural language

- Architecture diagrams can be created in seconds
- Diagrams open directly in the draw.io editor
- Diagrams remain fully editable

Example prompt:

“Create a 3-tier AWS architecture with Load Balancer, EC2 instances and an RDS database.”

Within seconds, the AI produces a diagram.

Prerequisites

Before configuring the MCP server, ensure the following are installed:

1. Node.js & npm

Required to run the MCP server via **npx**

Check installation:

```
node -v  
npm -v
```

2. Draw.io VS Code Extension

Install the unofficial draw.io integration:

Draw.io Integration by Henning Dieterichs

This extension allows you to:

- Open .drawio files inside VS Code

- Edit diagrams visually
- Render diagrams generated by AI tools

Step 1 — Configure the MCP Server in VS Code

To enable Draw.io diagram generation across all projects:

Open MCP User Configuration

1. Open Command Palette

Ctrl + Shift + P

2. Search for:

MCP: Open User Configuration

3. Add the following configuration to **mcp.json**

```
{
  "mcpServers": {
    "drawio": {
      "command": "npx",
      "args": ["-y", "@drawio/mcp"]
    }
  }
}
```

```
}  
}
```

if you face issue with this code please change to -

```
{  
  "Servers": {  
    "drawio": {  
      "command": "npx",  
      "args": ["-y", "@drawio/mcp"]  
    }  
  }  
}
```

Click on “Start” to start the server after adding this code .

Why `-y` is important

The `-y` flag allows `npx` to download and run the package without prompting for confirmation.

Verify the MCP Server

Restart VS Code.

Then check whether the server started.

Open Command Palette

```
Ctrl + Shift + P
```

Run:

```
MCP: List Servers
```

You should see:

```
drawio
```

If working correctly, VS Code logs will show something like:

```
Connection state: Running  
Discovered 3 tools
```

How to Generate a Diagram

Once the server is running, open **GitHub Copilot Chat**

Try prompts like:

Flowchart example

Create a draw.io flowchart for a user login process.

AWS architecture

Create a 3-tier AWS architecture diagram with a Load Balan

Organization chart

Convert this CSV into a draw.io org chart.

The AI will call MCP tools such as:

- create_diagram
- open_drawio_xml
- open_drawio_csv

Usually the response includes a **draw.io link**.

Click the link to open the diagram in the draw.io editor. (by default draw.io gets opened which do not contain the

architecture , *** Please Open Link ***)

Final Thoughts

The Draw.io MCP server is an excellent example of how AI assistants are evolving from text generators into development tools.

With the right setup, developers can:

- Generate architecture diagrams instantly
- Convert data into visualizations
- Design cloud architectures through prompts
- Maintain editable diagrams inside VS Code

As MCP tooling matures, diagram generation will likely become a core capability of AI-assisted development workflows.

AI

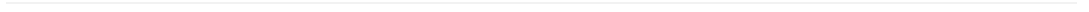
Drawio



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